STAFF REVIEW · H446 007 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. Because corresponding mantissa bit positions must represent the same scaled place values . Aligning the exponents puts both operands on the same scale before the mantissas are added.

2. 0.1875 Mantissa 01100000 = 0.75 ; exponent 1110 = −2 ; $0.75 \times 2^{-2} = 0.75 \div 4 = 0.1875$.

3. 01101101_2 6 → 0110 and D → 1101 .

4. +18 = 00010010 +29 = 00011101 −29 = 11100011 00010010 + 11100011 = 11110101 11110101 represents −11 . So 18 − 29 = 18 + (−29) = −11 .

I Do — source worked steps

1. Use common exponent 3 — Align both mantissas using sign-preserving right shifts. — CODE:
A: 0010100
B: 0000110

2. Invert and add 1 — CODE:
B: 0000110
invert → 1111001
add 1 → 1111010 — Negate the aligned mantissa, not the exponent.

3. Calculate A + (−B) — CODE:
0010100
+ 1111010
= 0001110 — The working difference is $14/32 \times 2^3$.

4. Shift left and reduce the exponent — CODE:
0001110×2^3
$\rightarrow 011100 \times 2^2$

Final: 011100 0010 — Check: 5 − 1.5 = 3.5 .

We Do — guided CHECK

1. Build the answer

2. Build the answer

3. Build the answer

4. Build the answer
Recognise the all-zero result and stop; it does not need further normalisation.

5. Build the answer

You Do — source mark schemes

1. Common exponent: 4 (0100). Align both mantissas with sign-preserving right shifts. Keep 1 extra low-order guard bit(s), and one extra leading sign bit. The working binary point stays before the 6 fractional bits.
CODE:
Aligned A: 00001001
Aligned B: 00100100
Invert B: 11011011
Add 1 to negate B: 11011100
A + (−B): 11100101
The working difference is -27/64 × 2 4 . Normalise and return to the requested field widths.
CODE:
Final: 100101 0011
Denary check: 2.25 − (9) = -6.75.
Common exponent 4 (1), exact aligned mantissas (1), negation of B (1), addition working (1), mantissa 100101 (1), exponent 0011 (1). Result −6.75.

2. Common exponent: 4 (0100). Align both mantissas with sign-preserving right shifts. Keep 0 extra low-order guard bit(s), and one extra leading sign bit. The working binary point stays before the 5 fractional bits.
CODE:
Aligned A: 0010010
Aligned B: 0001101
Invert B: 1110010
Add 1 to negate B: 1110011
A + (−B): 0000101
The working difference is 5/32 × 2 4 . Normalise and return to the requested field widths.
CODE:
Final: 010100 0010
Denary check: 9 − (6.5) = 2.5.
Common exponent 4 (1), aligned second mantissa 001101 (1), correct negation/addition (1), mantissa 010100 (1), exponent 0010 (1). Result 2.5.

3. Common exponent: 2 (0010). Align both mantissas with sign-preserving right shifts. Keep 0 extra low-order guard bit(s), and one extra leading sign bit. The working binary point stays before the 5 fractional bits.
CODE:
Aligned A: 0011000
Aligned B: 1110000
Invert B: 0001111
Add 1 to negate B: 0010000
A + (−B): 0101000
The working difference is 40/32 × 2 2 . Normalise and return to the requested field widths.
CODE:
Final: 010100 0011
Denary check: 3 − (-2) = 5.
Common exponent 2 (1), aligned second mantissa 110000 (1), negation/addition (1), final mantissa 010100 (1), exponent 0011 (1). Result 5.

4. a) 000000 0000 (1). b) Cancellation leaves redundant leading sign bits / a small mantissa (1); shifting left removes them, so the exponent must decrease to preserve the value (1). c) Discarding

a nonzero bit changes the operand, causing premature loss of precision (1).

Exit Ticket — source answer

1. [No safely extracted source model: staff must check]